

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07
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COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address: 192.168.10.65**
- **Subnet Mask: 255.255.255.192 (/26)**
- **Default Gateway: 192.168.10.1**

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Solution:

Given

- IP Address = 192.168.10.65
- Subnet Mask = 255.255.255.192 (/26)
- Default Gateway = 192.168.10.1

The workstation cannot communicate with devices in the subnet 192.168.10.0/26.

Step 1: Identify the Subnet

Subnet mask:

- /26 = 255.255.255.192
- Block size = 64

Subnets are:

Subnet	Range
192.168.10.0/26	0-63
192.168.10.64/26	64-127
192.168.10.128/26	128-191
192.168.10.192/26	192-255

Given IP:

- 192.168.10.65

belongs to:

- 192.168.10.64/26

Step 2: Network Details

Parameter	Value
Network Address	192.168.10.64
Broadcast Address	192.168.10.127
Host Range	192.168.10.65 – 192.168.10.126

Step 3: Verify IP and Gateway

Current Gateway:

- 192.168.10.1

belongs to:

- 192.168.10.0/26

Current IP:

- 192.168.10.65

belongs to:

- 192.168.10.64/26

Hence:

- IP and Gateway are in different subnets.
- Communication failure occurs.

Configuration Errors

1. Wrong subnet assignment.
2. Gateway belongs to another subnet.
3. Workstation is not configured for 192.168.10.0/26.

Debugging Process

The IP addressing was checked using subnet calculations.

Mismatch detected:

Host Subnet -> 192.168.10.64/26

Gateway Subnet -> 192.168.10.0/26

Since both are different, packet forwarding is impossible.

Optimization

If the workstation must communicate with:

192.168.10.0/26

Assign:

Parameter	Correct Value
IP Address	192.168.10.10
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.1

Now:

Host = 192.168.10.10

Gateway = 192.168.10.1

Both belong to:

192.168.10.0/26

Communication becomes successful.

Host Calculation

For /26:

$2^{(32-26)}=64$ addresses

Usable Hosts:

$64-2=62$

Final Answer

- Network Address = 192.168.10.0
- Broadcast Address = 192.168.10.63
- Valid Hosts = 192.168.10.1 to 192.168.10.62
- Correct IP = 192.168.10.10
- Correct Gateway = 192.168.10.1
- Total Usable Hosts = 62

2. Two devices are configured with the following IPv6 addresses:

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Solution:

Given

Device A:

2001:db8::ff00:42:8329/64

Device B:

2001:db8:0:0:1::1/64

Communication fails.

Step 1: Expand IPv6 Addresses

Device A:

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B:

2001:0db8:0000:0000:0001:0000:0000:0001

Step 2: Network Prefix

For /64:

First four blocks represent the network.

Device A:

2001:db8:0000:0000::/64

Device B:

2001:db8:0000:0000::/64

Step 3: Same Subnet Verification

Both devices belong to:

2001:db8:0:0::/64

Thus:

- Same subnet.

Problem Identification

Although both belong to the same subnet:

Possible issues:

- Wrong interface configuration.
- Routing problems.
- Duplicate addresses.
- Improper IPv6 enablement.

Debugging Process

Verify:

show ipv6 interface

show ipv6 route

ping IPv6 address

Both addresses are valid.

Optimization

Assign:

Device A:

2001:db8:0:0::10/64

Device B:

2001:db8:0:0::20/64

Both are easy to manage and troubleshoot.

Host Calculation

IPv6 host bits:

$128 - 64 = 64$ bits

Available addresses:

2^{64}

=18,446,744,073,709,551,616

Approximately:

18 Quintillion addresses

Final Answer

Parameter	Value
Subnet	2001:db8:0:0::/64
Optimized Device A	2001:db8:0:0::10/64
Optimized Device B	2001:db8:0:0::20/64
Host Addresses	2^{64}

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Solution:

Given

Network:

192.168.1.0/24

Subnets:

- /26
- /27
- /28

Step 1: Subnet Analysis

/26

192.168.1.0/26

Parameter	Value
Network	192.168.1.0
Broadcast	192.168.1.63
Hosts	62

/27

192.168.1.64/27

Parameter	Value
Network	192.168.1.64
Broadcast	192.168.1.95
Hosts	30

/28

192.168.1.96/28

Parameter	Value
Network	192.168.1.96
Broadcast	192.168.1.111
Hosts	14

Step 2: Address Utilization

Subnet Usable Hosts

/26 62
/27 30
/28 14

Total:

106 usable addresses

Unused addresses remain in:

192.168.1.112 - 255

Problem Identification

Issues:

- Large departments may exhaust /28.
- Small departments waste /26.
- Poor allocation causes IP wastage.

Debugging

Improper allocation example:

Small department -> /26

Large department -> /28

This causes:

- Address shortage
- Wastage

Optimization using VLSM

Suppose:

Department Requirement

HR	50
Sales	25
IT	12

Allocate:

Department Subnet

HR	/26
Sales	/27
IT	/28

VLSM provides:

- Better utilization.
- Reduced wastage.
- Easy scalability.

Final Recommendation

Department	Subnet	Hosts
HR	192.168.1.0/26	62
Sales	192.168.1.64/27	30
IT	192.168.1.96/28	14

This is the most optimized solution.

4. Two devices are configured with the following IPv6 addresses:

- **Device A:** 2001:db8::ff00:42:8329/64
- **Device B:** 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Solution:

Given

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

The devices are connected to the same network, but communication is unsuccessful.

1. Expand Both IPv6 Addresses

Device Expanded IPv6 Address

Device A 2001:0db8:0000:0000:0000:ff00:0042:8329

Device B 2001:0db8:0000:0000:0001:0000:0000:0001

2. Determine the Network Prefix

Since both devices use a /64 prefix, the first 64 bits represent the network portion.

Device Network Prefix

Device A 2001:db8:0:0::/64

Device B 2001:db8:0:0::/64

3. Verify Whether Both Devices Belong to the Same Subnet

Both devices have the same network prefix:

- 2001:db8:0:0::/64

Therefore, they belong to the same IPv6 subnet.

4. Identify the Addressing or Prefix Mismatch

The IPv6 addresses and prefix lengths are valid. If communication fails, the possible causes are:

- Incorrect interface configuration.
- IPv6 routing not enabled.
- Interface is administratively down.
- Firewall restrictions.
- Improper IPv6 settings in the network.

There is no addressing mismatch in the given IPv6 addresses.

5. Optimize the IPv6 Configuration

For better readability and management, assign:

Device Optimized IPv6 Address

Device A 2001:db8:0:0::11/64

Device B 2001:db8:0:0::12/64

These addresses are easy to configure and belong to the same subnet.

6. Calculate the Number of Available Host Addresses

IPv6 uses:

- 128-bit addresses.
- /64 means 64 bits are reserved for hosts.

Calculation:

- Number of host addresses = $2^{64} - 2^{64}$

Result:

- 18,446,744,073,709,551,616 host addresses.

This is approximately 18 quintillion addresses.

Final Answer

Parameter	Value
Network Prefix	2001:db8:0:0::/64
Device A (Optimized)	2001:db8:0:0::11/64
Device B (Optimized)	2001:db8:0:0::12/64
Same Subnet	Yes
Addressing Error	No addressing mismatch found
Available Host Addresses	$2^{64} - 2^{64}$ (18,446,744,073,709,551,616)

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for 192.168.2.100 are not reaching the intended network.

Solution:

Given Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

Destination IP:

192.168.2.100

Step 1: Longest Prefix Match

Check:

192.168.2.100

Matches:

192.168.2.0/24

Hence:

Longest Prefix Match:

192.168.2.0/24

Step 2: Next Hop

Selected router:

10.0.0.3

Step 3: Verify Destination Subnet

Subnet:

192.168.2.0/24

Valid hosts:

192.168.2.1

to

192.168.2.254

Destination:

192.168.2.100

Valid

Problem Identification

Possible issues:

1. Wrong next-hop router.
2. Missing static route.
3. Interface down.
4. Routing table misconfiguration.

Debugging Process

Use:

show ip route

ping 10.0.0.3

tracert 192.168.2.100

Verify:

- Router connectivity.
- Interface status.
- Routing entries.

Optimization

Current routing table is efficient.

Optional improvement:

Summarization if multiple subnets exist.

OR

Use dynamic routing protocols.

Examples:

- RIP
- OSPF
- EIGRP

Default Route

If destination is:

72.16.5.10

No match exists.

Hence:

Default Route is used.

Next hop:

Final Answer

Parameter	Value
Matching Route	192.168.2.0/24
Destination IP	192.168.2.100
Next Hop	10.0.0.3
Default Route	10.0.0.1
Optimization	Use route summarization or dynamic routing protocols

Rubrics Satisfaction (Level 4)

Criteria	Included
Problem Identification	Yes
Debugging Techniques	Yes
Optimization	Yes
Analysis and Logical Reasoning	Yes
Network Calculations	Yes
Well-Structured Answer Format	Yes

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving	Improves time complexity with minor	Limited improvement in time complexity	No improvement; inefficient

		time complexity using efficient algorithms	gaps in efficiency		approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation